Parking Garage

Scenario C, Round 1 - 60 minutes, teams of 4 to 5

Your task

Your team builds a guidance system for a small parking garage. One sensor watches the entry and one watches the exit. The system counts the cars inside and shows a traffic light. Green means free spaces, yellow means almost full and red means full. When the garage is full, new cars are refused.

In this round your team works on one shared program. Apart from the roles, organize yourselves freely.

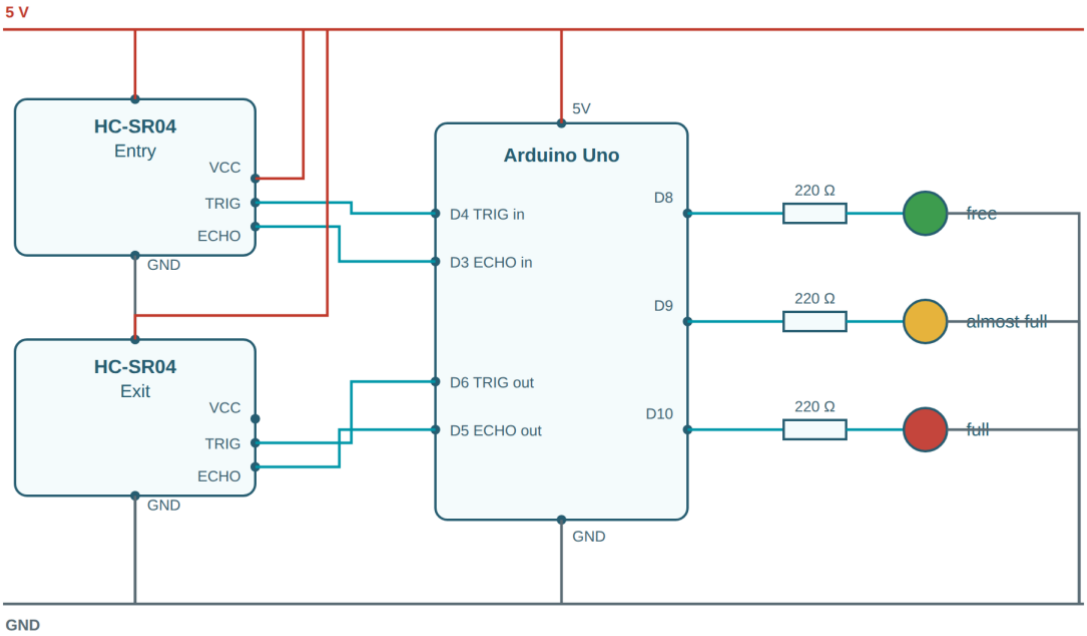
What the system must do

- Each car that enters is counted once, even if it stays in front of the sensor.
- Each car that leaves is subtracted once, even if it stays in front of the sensor.
- The count never goes below 0 and never above the capacity of 10.
- Green below 8 cars, yellow from 8 cars, red at 10 cars. Only one LED is on at a time.
- Every change appears in the Serial Monitor.

Parts

Part	Quantity
Arduino Uno and breadboard	1
HC-SR04 ultrasonic sensor	2
LED	3
220 Ohm resistor	3
Jumper wires	about 20

Wiring



Use the pins shown in the diagram. Every LED needs a 220 Ohm resistor in series.

The code

You can build the circuit on a real board or in TinkerCad Circuits. In TinkerCad, open the code editor, switch it to Text and paste the code. Fill in every blank marked ___ before you start the program. The code you get is not fully correct. When the system does not do what the list above asks, find the cause and fix it.

Blank	What to fill in
Capacity	The number of cars the garage can hold. It appears in the entry check and in the red LED line.
Almost full	The number of cars from which the garage counts as almost full. It appears in the green and the yellow LED line.

```
/*
 * Scenario C  Parking garage
 * Round 1  One shared program
 * Fill in every ___ before you upload.
 */

int cars = 0;
bool entryBusy = false, exitBusy = false;
unsigned char entryNear = 0, entryFar = 0, exitNear = 0, exitFar = 0;
unsigned long lastTick = 0;

long measureDistanceCm(uint8_t trigPin, uint8_t echoPin) {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  long duration = pulseIn(echoPin, HIGH, 30000UL);
  if (duration == 0) return 400;
```

```
  return duration / 58;
}

void setup() {
  pinMode(4, OUTPUT);  pinMode(3, INPUT);   // entry sensor
  pinMode(6, OUTPUT);  pinMode(5, INPUT);   // exit sensor
  pinMode(8, OUTPUT);  pinMode(9, OUTPUT);  pinMode(10, OUTPUT);
  Serial.begin(9600);
  Serial.print("Parking garage ready. Capacity: ");
  Serial.println(10);
  digitalWrite(8, HIGH);  digitalWrite(9, LOW);  digitalWrite(10, LOW);
}

void loop() {
  if (millis() - lastTick < 60) return;
  lastTick = millis();

  // entry lane, exit lane, counting and LEDs, all interleaved
  long dIn = measureDistanceCm(4, 3);
  if (dIn < 15) { entryNear++; entryFar = 0; } else { entryFar++; entryNear = 0; }
  if (!entryBusy && entryNear >= 2) {
    entryBusy = true;
    if (cars < ___) {                // capacity of the garage
      cars++;
      Serial.print("Car in.  Count: ");
      Serial.println(cars);
    } else {
      Serial.println("Entry refused: garage FULL.");
    }
  }
  if (entryBusy && entryNear >= 2) entryBusy = false;

  long dOut = measureDistanceCm(6, 5);
  if (dOut < 15) { exitNear++; exitFar = 0; } else { exitFar++; exitNear = 0; }
  if (!exitBusy && exitNear >= 2) {
    exitBusy = true;
    if (cars >= 0) {
      cars--;
      Serial.print("Car out. Count: ");
      Serial.println(cars);
    } else {
      Serial.println("Exit ignored: garage already empty.");
    }
  }
  if (exitBusy && exitNear >= 2) exitBusy = false;

  digitalWrite(8,  cars < ___ ? HIGH : LOW);    // green
  digitalWrite(9,  cars >= ___ ? HIGH : LOW);   // yellow
  digitalWrite(10, cars >= ___ ? HIGH : LOW);   // red
}
```

How you work

Every member of the team takes one role.

Role	Responsibility	Works closely with
Team lead	Keeps time and signs off the result.	Everyone
Detection developer	Takes care of the entry and exit sensors in the shared program.	Integrator
Counting developer	Takes care of the count and its limits in the shared program.	Display developer, integrator
Display developer	Takes care of the traffic light in the shared program.	Counting developer

Role	Responsibility	Works closely with
Integrator	Keeps the shared program running and tests the whole system.	Everyone

In a team of four, the team lead also takes the role of integrator.

There is one shared program and no fixed rules for who changes which part of it.

You have 60 minutes.

After the round

Take about 10 minutes as a team and talk about these questions.

- When something did not work, how did you find the cause?
- Could several people work on the code at the same time?
- How did you split the work?
- What would you change if you did this again?